

Finite Element Method

Finite Element Approximation: 1-D Problems II

MECH4103 — Spring 2026

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- Expressions for heat conduction and diffusion are obtained by replacing fields and parameters using the conversion table from Lecture 3.
- Terminology of the discrete matrices:

Matrix	Elasticity	Diffusion	Heat conduction
K	Stiffness	Diffusivity	Conductance
f	Force	Flux	Flux
d	Displacement	Concentration	Temperature

- The element matrices are:

$$\mathbf{K}^e = \int_{\Omega^e} (\mathbf{B}^e)^T A^e \kappa^e \mathbf{B}^e dx,$$

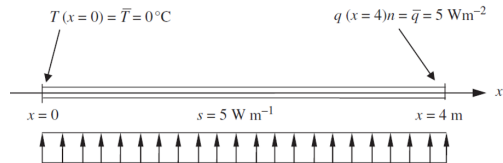
$$\mathbf{f}^e = \underbrace{\int_{\Omega^e} (\mathbf{N}^e)^T f dx}_{\mathbf{f}_{\Omega}^e} + \underbrace{((\mathbf{N}^e)^T A^e \bar{\Phi})|_{\Gamma_{\Phi}^e}}_{\mathbf{f}_{\Gamma}^e} \quad (1)$$

Field/param.	Elasticity	Heat cond.	Diffusion
θ	u	T	c
κ	E	k	k
f	b	s	s
$\bar{\Phi}$	$\bar{t}$	$-\bar{q}$	$-\bar{q}$

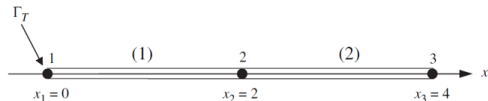
- A bar with uniformly distributed heat source $s = 5 \text{ W m}^{-1}$, cross-sectional area $A = 0.1 \text{ m}^2$, thermal conductivity $k = 2 \text{ W }^\circ\text{C}^{-1} \text{ m}^{-1}$, length $l = 4 \text{ m}$.
- Boundary conditions:

$$T(0) = \bar{T} = 0^\circ\text{C}, \quad \bar{q}(x = 4) = 5 \text{ W m}^{-2}.$$

- Divide the domain into **two linear two-node elements** and solve by FEM.



- Number the nodes starting from Γ_T :



$$x_1 = 0, \quad x_2 = 2, \quad x_3 = 4.$$

Element conductance matrix

- The two-node element gives (replacing E by k):

$$\mathbf{K}^e = \int_{\Omega^e} (\mathbf{B}^e)^T A^e k^e \mathbf{B}^e dx = \frac{A^e k^e}{l^e} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

- For element 1: $x_1^{(1)} = 0$, $x_2^{(1)} = 2$, $l^{(1)} = 2$, $(Ak)^{(1)} = 0.2$:

$$\mathbf{K}^{(1)} = \frac{0.2}{2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 0.1 & -0.1 \\ -0.1 & 0.1 \end{bmatrix}$$

- For element 2:

$$\mathbf{K}^{(2)} = \begin{bmatrix} 0.1 & -0.1 \\ -0.1 & 0.1 \end{bmatrix}$$

- The global conductance matrix is assembled as:

$$\mathbf{K} = \sum_{e=1}^{n_{el}} (\mathbf{L}^e)^T \mathbf{K}^e \mathbf{L}^e = (\mathbf{L}^{(1)})^T \mathbf{K}^{(1)} \mathbf{L}^{(1)} + (\mathbf{L}^{(2)})^T \mathbf{K}^{(2)} \mathbf{L}^{(2)}$$

- The gather matrices for the two elements:

$$\mathbf{d}^{(1)} = \begin{bmatrix} T_1^{(1)} \\ T_2^{(1)} \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}}_{\mathbf{L}^{(1)}} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \end{bmatrix} = \mathbf{L}^{(1)} \mathbf{d}$$

$$\mathbf{d}^{(2)} = \begin{bmatrix} T_1^{(2)} \\ T_2^{(2)} \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\mathbf{L}^{(2)}} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \end{bmatrix} = \mathbf{L}^{(2)} \mathbf{d}$$

- The scatter of the conductance matrices gives:

$$\tilde{\mathbf{K}}^{(1)} = (\mathbf{L}^{(1)})^T \mathbf{K}^{(1)} \mathbf{L}^{(1)} = \begin{bmatrix} 0.1 & -0.1 & 0 \\ -0.1 & 0.1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\tilde{\mathbf{K}}^{(2)} = (\mathbf{L}^{(2)})^T \mathbf{K}^{(2)} \mathbf{L}^{(2)} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0.1 & -0.1 \\ 0 & -0.1 & 0.1 \end{bmatrix}$$

- The total conductance matrix is the sum of the scattered element matrices:

$$\mathbf{K} = \tilde{\mathbf{K}}^{(1)} + \tilde{\mathbf{K}}^{(2)} = \begin{bmatrix} 0.1 & -0.1 & 0 \\ -0.1 & 0.2 & -0.1 \\ 0 & -0.1 & 0.1 \end{bmatrix}$$

- Direct assembly** gives the same result:

$$\mathbf{K}^{(1)} = \begin{bmatrix} 0.1 & -0.1 \\ -0.1 & 0.1 \end{bmatrix} \begin{matrix} [1] \\ [2] \end{matrix} \quad \begin{matrix} [1] & [2] \end{matrix}$$

$$\mathbf{K}^{(2)} = \begin{bmatrix} 0.1 & -0.1 \\ -0.1 & 0.1 \end{bmatrix} \begin{matrix} [2] \\ [3] \end{matrix} \quad \begin{matrix} [2] & [3] \end{matrix}$$

$$\mathbf{K} = \begin{bmatrix} 0.1 & -0.1 & 0 \\ -0.1 & 0.2 & -0.1 \\ 0 & -0.1 & 0.1 \end{bmatrix} \begin{matrix} [1] \\ [2] \\ [3] \end{matrix}$$

Result

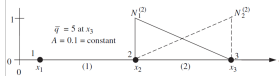
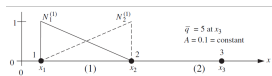
Both procedures yield the identical global conductance matrix.

Boundary flux matrix

- Only shape functions nonzero on Γ_q contribute to the boundary flux. Only elements on the natural boundary are considered.

- The element boundary flux matrix (replacing $\bar{t}$ by $-\bar{q}$):

$$\mathbf{f}_\Gamma^e = -((\mathbf{N}^e)^T A^e \cdot \bar{q})|_{\Gamma_q^e} = -\mathbf{N}^{eT}(x_3) \times 0.1 \times 5 = -0.5 \mathbf{N}^{eT}(x_3)$$



- For element 1 ($N_1^{(1)}(x_3) = 0$, $N_2^{(1)}(x_3) = 0$):

$$\mathbf{f}_\Gamma^{(1)} = -0.5 \begin{bmatrix} N_1^{(1)}(x_3) \\ N_2^{(1)}(x_3) \end{bmatrix} = -0.5 \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

- For element 2 ($N_1^{(2)}(x_3) = 0$, $N_2^{(2)}(x_3) = 1$):

$$\mathbf{f}_\Gamma^{(2)} = -0.5 \begin{bmatrix} N_1^{(2)}(x_3) \\ N_2^{(2)}(x_3) \end{bmatrix} = -0.5 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -0.5 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

- The scatter (direct assembly) gives the global boundary flux matrix:

$$\mathbf{f}_\Gamma = \sum_{e=1}^2 (\mathbf{L}^e)^T \mathbf{f}_\Gamma^e = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ -0.5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -0.5 \end{bmatrix} \begin{matrix} [1] \\ [2] \\ [3] \end{matrix}$$

- The element source flux matrix is:

$$\mathbf{f}_{\Omega}^e = \int_{x_1^e}^{x_{n_{\text{en}}}^e} (\mathbf{N}^e)^T s \, dx = \frac{l^e}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} s_1 \\ s_2 \end{bmatrix}$$

- Since $s_1 = s_2 = s$ (uniform source), this reduces to:

$$\mathbf{f}_{\Omega}^e = \frac{l^e s}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

- With $l^{(1)} = l^{(2)} = 2$ and $s = 5$:

$$\mathbf{f}_{\Omega}^{(1)} = \mathbf{f}_{\Omega}^{(2)} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$$

- Assembling using direct assembly:

$$\mathbf{f}_{\Omega} = \sum_{e=1}^2 (\mathbf{L}^e)^T \mathbf{f}_{\Omega}^e = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 5 \\ 5 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 5 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \\ 5 \end{bmatrix}$$

- The global system of equations is:

$$\begin{bmatrix} 0.1 & -0.1 & 0 \\ -0.1 & 0.2 & -0.1 \\ 0 & -0.1 & 0.1 \end{bmatrix} \begin{bmatrix} 0 \\ T_2 \\ T_3 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \\ 5 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -0.5 \end{bmatrix} + \begin{bmatrix} r_1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} r_1 + 5 \\ 10 \\ 4.5 \end{bmatrix}$$

- Since node 1 is on the essential boundary ($T_1 = \bar{T} = 0$), partition after the first row:

$$\begin{bmatrix} 0.2 & -0.1 \\ -0.1 & 0.1 \end{bmatrix} \begin{bmatrix} T_2 \\ T_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 4.5 \end{bmatrix} \Rightarrow \begin{bmatrix} T_2 \\ T_3 \end{bmatrix} = \begin{bmatrix} 145 \\ 190 \end{bmatrix} \text{ } ^\circ\text{C}$$

Postprocessing — temperature gradient

$$\frac{dT^{(1)}}{dx} = \mathbf{B}^{(1)} \mathbf{L}^{(1)} \mathbf{d} = \frac{1}{2} \begin{bmatrix} -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 145 \\ 190 \end{bmatrix} = 72.5 \quad \frac{dT^{(2)}}{dx} = \mathbf{B}^{(2)} \mathbf{L}^{(2)} \mathbf{d} = \frac{1}{2} \begin{bmatrix} -1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 145 \\ 190 \end{bmatrix} = 22.5$$

Exact analytical solution

- The governing equation with $Ak = 0.2$:

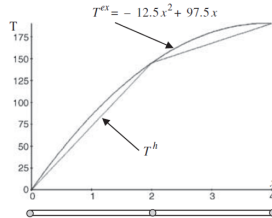
$$\frac{d}{dx} \left(0.2 \frac{dT}{dx} \right) + 5 = 0 \quad \Rightarrow \quad \frac{d^2 T}{dx^2} = -25$$

- BCs: $T(0) = 0$ and $\bar{q}(4) = -k \frac{dT}{dx} n|_{x=4} = 5 \Rightarrow \frac{dT}{dx}(4) = -2.5$.
- Integrating twice:

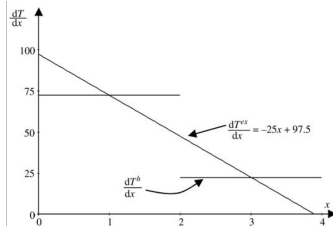
$$\begin{aligned} \frac{dT}{dx} &= -25x + c_1, \quad c_1 = 97.5, \\ T &= -12.5x^2 + 97.5x + c_2, \quad c_2 = 0. \end{aligned}$$

- Exact solution:

$$T^{\text{ex}} = -12.5x^2 + 97.5x, \quad \frac{dT^{\text{ex}}}{dx} = -25x + 97.5.$$



Comparison of the exact and finite element solutions of temperature.



Comparison of the exact and finite element solutions of temperature gradient.

- The figures compare the FEM solution with the exact solution.
- The derivative of the temperature (and hence the flux) is a C^{-1} function in the FEM — it is **discontinuous between elements** (kinks at shared nodes).
- The **essential boundary condition** $T(0) = 0$ is satisfied exactly. In FEM, essential boundary conditions are *always* satisfied exactly.

FEM vs exact — temperatures

- FEM: $T_1 = 0$, $T_2 = 145$, $T_3 = 190$ °C
- Exact at $x = 2$: $T^{\text{ex}}(2) = -50 + 195 = 145$ °C ✓
- Exact at $x = 4$: $T^{\text{ex}}(4) = -200 + 390 = 190$ °C ✓

Temperature gradient

- FEM elem. 1: $dT/dx = 72.5$ (constant)
 - FEM elem. 2: $dT/dx = 22.5$ (constant)
 - Exact: $-25x + 97.5$ (linear, continuous)
 - FEM is C^{-1} in derivative; exact is C^∞
- The nodal temperatures are **exact** in this case, because the exact solution is quadratic and the FEM stiffness matrix integrates it exactly via the two-point Gauss rule.

Next: Convergence of the FEM

Thanks for your attention!